

GOAD - Game Of Active Directory

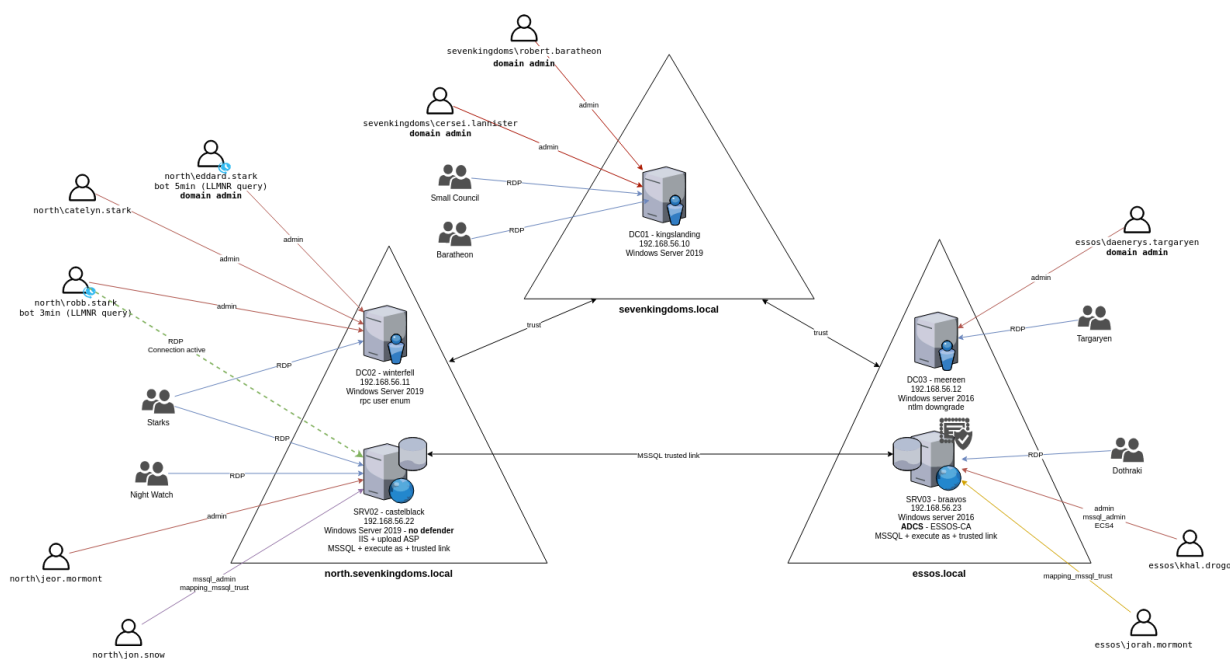
orange-cyberdefense.github.io/GOAD/labs/GOAD

mayfly



GOAD

GOAD is the first and main lab of this project. It contains 3 domains and 2 forest.



Servers

This lab is actually composed of five virtual machines:

domain sevenkingdoms.local

kingslanding : DC01 running on Windows Server 2019 (with windefender enabled by default)

domain north.sevenkingdoms.local

- **winterfell** : DC02 running on Windows Server 2019 (with windefender enabled by default)
- **castelblack** : SRV02 running on Windows Server 2019 (with windefender **disabled** by default)

domain essos.local

- **meereen** : DC03 running on Windows Server 2016 (with windefender enabled by default)
- **braavos** : SRV03 running on Windows Server 2016 (with windefender enabled by default)

WRITEUP

All the writeups of the Game Of Active Directory lab are available on mayfly's blog :

Computers Users and group permissions

- **SEVENKINGDOMS / sevenkingdoms.local**

DC01 : kingslanding.sevenkingdoms.local (Windows Server 2019)
(SEVENKINGDOMS DC)

- Admins : robert.baratheon (U), cersei.lannister (U)
- RDP: Small Council (G)

- **NORTH / north.sevenkingdoms.local**

- DC02 : winterfell.north.sevenkingdoms.local (Windows Server 2019) (NORTH DC)

- Admins : eddard.stark (U), catelyn.stark (U), robb.stark (U)
- RDP: Stark(G)

- SRV02 : castelblack.essos.local (Windows Server 2019) (IIS, MSSQL, SMB share)

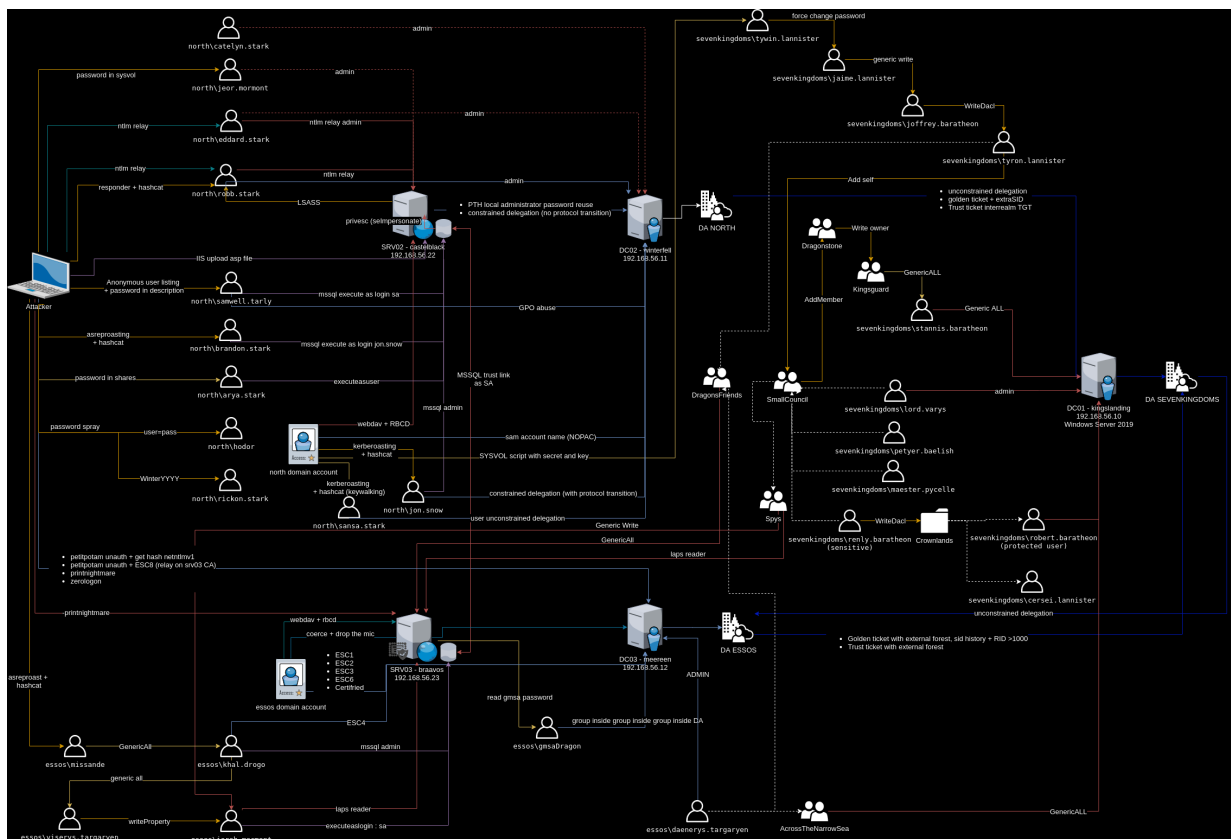
- Admins: jeor.mormont (U)
- RDP: Night Watch (G), Mormont (G), Stark (G)
- IIS : allow asp upload, run as NT Authority/network
- MSSQL:
 - admin : jon.snow
 - impersonate :
 - execute as login : samwel.tarlly -> sa
 - execute as user : arya.stark -> dbo
 - link :
 - to braavos : jon.snow -> sa

- **ESSOS / `essos.local`**

- DC03 : meereen.essos.local (Windows Server 2016) (ESSOS DC)
 - Admins: daenerys.targaryen (U)
 - RDP: Targaryen (G)
- SRV03 : braavos.essos.local (Windows Server 2016) (MSSQL, SMB share)
 - Admins: khal.drogo (U)
 - RDP: Dothraki (G)
 - MSSQL :
 - admin : khal.drogo
 - impersonate :
execute as login : jorah.mormont -> sa
 - link:
to castelblack: jorah.mormont -> sa

Users/Groups and associated scenarios

Graph of some scenarios is available here :



NORTH.SEVENKINGDOMS.LOCAL

- STARKS: RDP on WINTERFELL AND CASTELBLACK
 - arya.stark: Execute as user on mssql, pass on all share
 - eddard.stark: DOMAIN ADMIN NORTH/ (bot 5min) LLMRN request to do NTLM relay with responder
 - catelyn.stark:
 - robb.stark: bot (3min) RESPONDER LLMR / lsass present user
 - sansa.stark: keywalking password / unconstrained delegation
 - brandon.stark: ASREP_ROASTING
 - rickon.stark: pass spray WinterYYYY
 - jon.snow: mssql admin / KERBEROASTING / mssql trusted link
 - hodor: PASSWORD SPRAY (user=password)
- NIGHT WATCH: RDP on CASTELBLACK
 - samwell.tarly: Password in ldap description / mssql execute as login GPO abuse (Edit Settings on "STARKWALLPAPER" GPO)
 - jon.snow: (see stark)
 - jeor.mormont: (see mormont)
- MORMONT: RDP on CASTELBLACK
 - jeor.mormont: Admin castelblack, pass in sysvol script
- AcrossTheSea : cross forest group

SEVENKINGDOMS.LOCAL

- LANISTERS
 - tywin.lannister: ACE forcechange password on jaime.lanister, password on sysvol cyphered
 - jaime.lannister: ACE genericwrite-on-user joffrey.baratheon
 - tyron.lannister: ACE self membership on small council
 - cersei.lannister: DOMAIN ADMIN SEVENKINGDOMS
- BARATHEON: RDP on KINGSLANDING
 - robert.baratheon: DOMAIN ADMIN SEVENKINGDOMS, protected user
 - joffrey.baratheon: ACE Write DACL on tyron.lannister
 - renly.baratheon: WriteDACL on container, sensitive user
 - stannis.baratheon: ACE genericall-on-computer kingslanding
- SMALL COUNCIL : ACE add Member to group dragon stone / RDP on KINGSLANDING
 - petyer.baelish:
 - lord.varys: ACE genericall-on-group Domain Admins and sdholder
 - maester.pycelle:
- DRAGONSTONE : ACE Write Owner on group KINGSGUARD
- KINGSGUARD : ACE generic all on user stannis.baratheon
- AccorsTheNarrowSea: cross forest group

ESSOS.LOCAL

- TARGERYEN
 - missande : ASREP roasting, generic all on khal
 - daenerys.targaryen: DOMAIN ADMIN ESSOS
 - viserys.targaryen: ACE write property on jorah.mormont
 - jorah.mormont: mssql execute as login / mssql trusted link / Read LAPS Password
- DOTHRAKI
 - khal.drogo: mssql admin / GenericAll on viserys (shadow credentials) / GenericAll on ECS4
- DragonsFriends: cross forest group
- Spys: cross forest group / Read LAPS password / ACL generic all jorah.mormont